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4. Markov's inequality states that for any non-negative random variable X and any $a > 0$: $P(X \geq a) \leq \frac{E[X]}{a}$, where $E[X]$ is the expected value (average) of the random variable. Note that you do not need to have seen the Markov Inequality or its proof to answer this question. [By Sadra Sabouri Halestani]
- (a) Apply Markov's inequality to the random variable mean squared error on the test set $(y_0 - \hat{f}(x_0))^2$ and use the Bias-variance trade-off equation to derive an upper bound for the probability of squared error on the test set, $(y_0 - \hat{f}(x_0))^2$, being more than a given threshold, δ , based on model bias and variance and $\text{Var}(\epsilon)$.
- (b) Using the result in 4a, for a fixed δ , what models have a smaller probability of having large squared errors? (Assume that the mean squared error on the test set is less than δ)

- (c) Assume that ϵ is distributed according to a zero-mean normal with a variance of 1.0, $\epsilon \sim N(0, 1.0)$, Compare the following models by average squared error and upper bound probability of squared error for $\delta = 1$, using the above results:
- $\text{Var}(\hat{f}_1) = 2$, $\text{Bias}(\hat{f}_1) = -0.1$
 - $\text{Var}(\hat{f}_2) = 0.1$, $\text{Bias}(\hat{f}_2) = 2$
- (d) Repeat 4c with $\delta = 10$.

Prove that $\hat{Y} = E[Y|X = x]$

minimizes the MSE at each x

$$E[(Y - \hat{Y})^2 | X = x]$$

and therefore the total MSE

$$E \left[(Y - \hat{Y})^2 \right]$$